Dharaneshwar K

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Summary

Machine Learning Engineer / AI Researcher with hands-on experience building production-grade ML pipelines, ensemble models, and transformer-based systems at scale; first-author IEEE publication in survival-analysis-driven predictive modeling.

Education

Amrita Vishwa Vidyapeetham, Coimbatore — Jul 2026 – May 2028 (Expected)
M.Tech in Artificial Intelligence Engineering
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Data Mining

Saranathan College of Engineering, Trichy — Sep 2022 – Apr 2026
B.Tech in Artificial Intelligence and Data Science — ***CGPA: 7.72/10***
Relevant Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Machine Learning, Deep Learning, NLP, Computer Vision

Professional Experience

National Institute of Technology, Trichy — Jun 2025 - Sept 2025
Machine Learning Research Intern

- Researched survival analysis and ensemble learning techniques for workforce analytics under faculty guidance at NIT Trichy's ML & Data Analytics Lab.
- Built an end-to-end ML pipeline (preprocessing, CCA feature engineering, SMOTE balancing, model training/evaluation) in Python using Scikit-learn and XGBoost, improving predictive accuracy for employee attrition forecasting.

Skills

Languages: Python, Java, C, SQL
Frameworks & Libraries: TensorFlow, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, OpenCV, Keras, Matplotlib
Tools & Technologies: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, AWS
Domains: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Survival Analysis, Anomaly Detection, Transformers

Projects

- **Hybrid Survival Ensemble Model for Employee Attrition Prediction (2026):** Engineered a hybrid ensemble (RSF, CoxPH, XGBoost) with CCA-based feature engineering and SMOTE balancing, achieving 0.95 C-index and 0.97 AUROC on 7K+ records – outperforming CoxPH and WTTE-RNN baselines; published as first author at IEEE ICITIIT 2026.
- **Satellite Telemetry Anomaly Detection using TranAD (2026):** Built an unsupervised anomaly detection system for 103-dimensional ESA GOCE satellite telemetry using a transformer-based (TranAD) architecture with POT thresholding, from raw sensor preprocessing to detection pipeline. *Stack: Python, PyTorch, Transformers, Pandas, NumPy, Scikit-learn.*

Publications

- **Hybrid Survival Ensemble Model for Employee Attrition Prediction – 2026 IEEE ICITIIT**, IIIT Kottayam. DOI: 10.1109/ICITIIT68860.2026.11499599.
- **Markerless Autonomous UAV Landing Using Vision Transformers for Risk-Aware Terrain Segmentation (2026)** – Abstract, ICAAN Conference Proceedings.

Certifications

- AWS Cloud Quest — *Mar 2026*
- Data Analytics using Excel – SWAYAM (360DigiTMG) — *Sep 2025*
- Data Visualization – Tata Forage — *Jul 2025*
- Data Analytics – Deloitte Forage — *Jul 2025*
- Business Intelligence – Infosys Springboard — *Apr 2025*